Exp. No: 13 Date:	PERFORMANCE EVALUATION OF DENSE WDM MULTIPLEXING AND DEMULTIPLEXING USING OPTISYSTEM

OBJECTIVES

1. To design and simulate a Dense Wavelength Division Multiplexing (DWDM) system in OptiSystem and analyze its performance under different test conditions.
2. To analyze the effect of wavelength (channel) spacing, fiber length (loop count) and channel power on BER, Q-factor, OSNR and crosstalk.
3. To compare modulation formats (NRZ, RZ, QPSK) and determine the optimal optical amplifier (EDFA) placement for long-haul DWDM transmission.

SOFTWARE REQUIRED

• OptiSystem

I/P PARAMETERS

S.No.	Parameter	Value
1	Laser Frequency (ITU grid)	193.1 THz, 193.2 THz, …
2	Wavelength (Frequency) Spacing	100 GHz, 50 GHz, 25 GHz
3	Modulation Format	NRZ, RZ, QPSK
4	Channel Power	0 dBm to 10 dBm
5	Optical Fiber	Single-mode fiber (SMF)
6	Fiber Attenuation / Dispersion	0.2 dB/km / 16 ps/nm/km
7	Fiber Length (Loop Count)	50 km, 250 km, 500 km (1, 5, 10 loops)
8	Optical Amplifier	EDFA / Raman, noise figure 5 dB
9	Amplifier Spacing	50 km, 100 km, 150 km
10	Receiver	PIN / APD photodetector

O/P PARAMETERS

Test Case 1: Wavelength Spacing Variation

Spacing	Min BER	Max Q-Factor
100 GHz		
50 GHz		
25 GHz		

Test Case 2: Transmission Distance Variation

Loop Count (Length)	Min BER	Max Q-Factor
1 (50 km)		
5 (250 km)		
10 (500 km)		

Test Case 3: Modulation Format

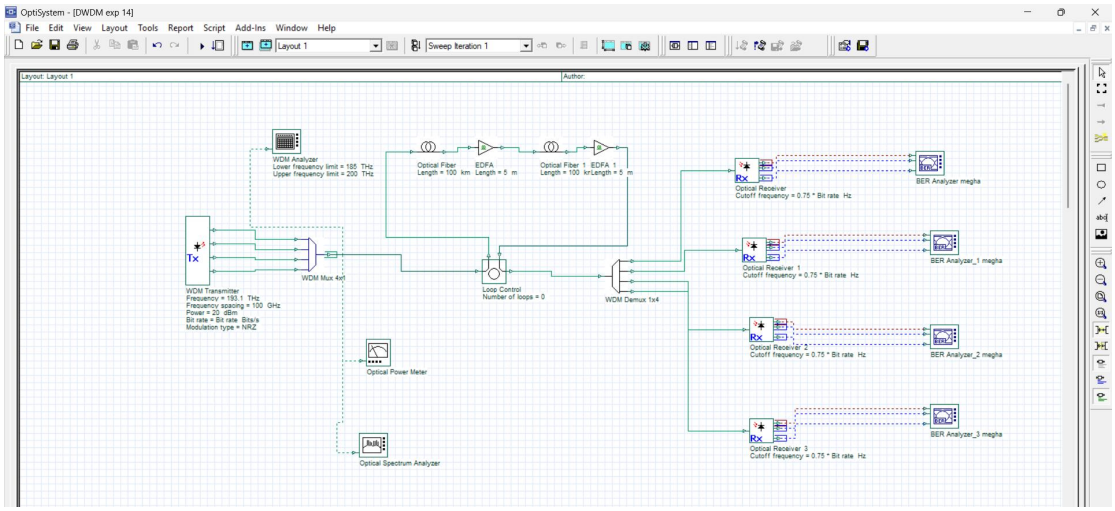
Format	Min BER	Max Q-Factor
NRZ		
RZ		
QPSK		

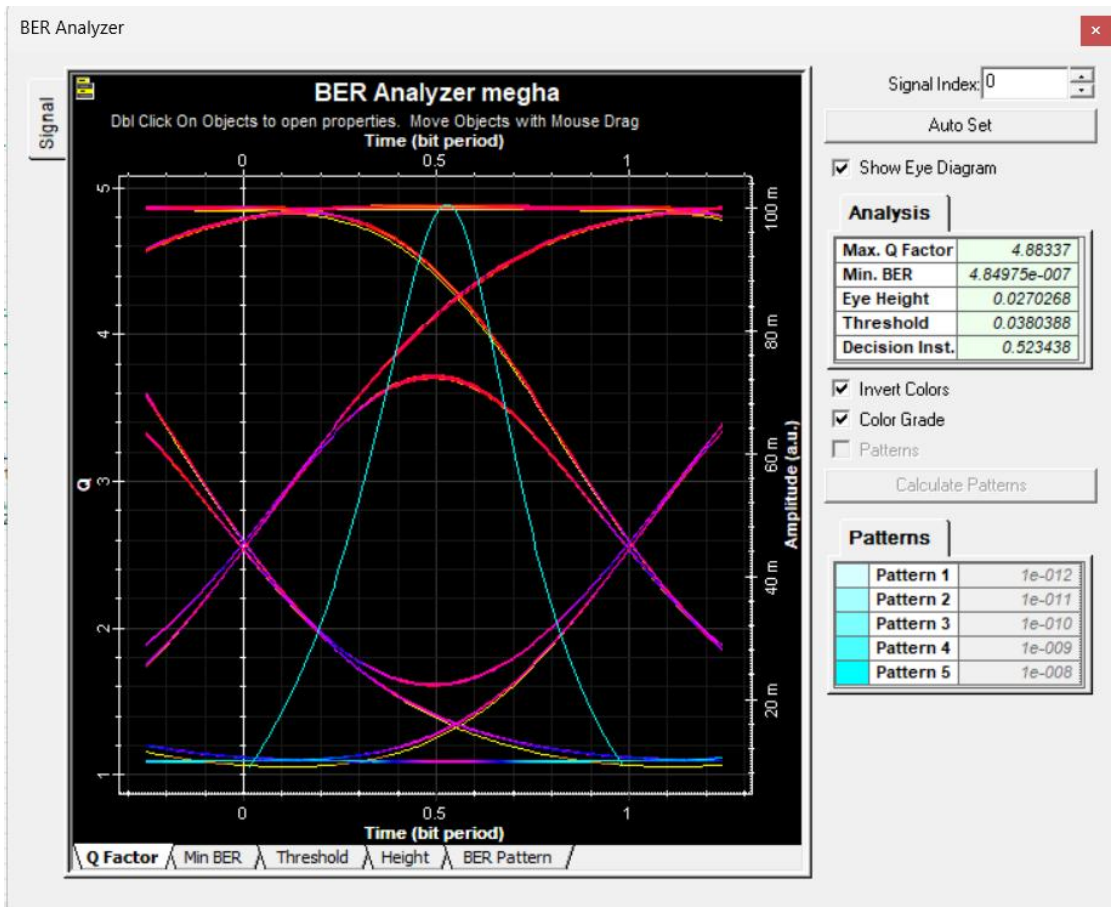
Test Case 4: Amplifier Placement

EDFA Spacing	Min BER	Max Q-Factor
50 km		
100 km		
150 km		

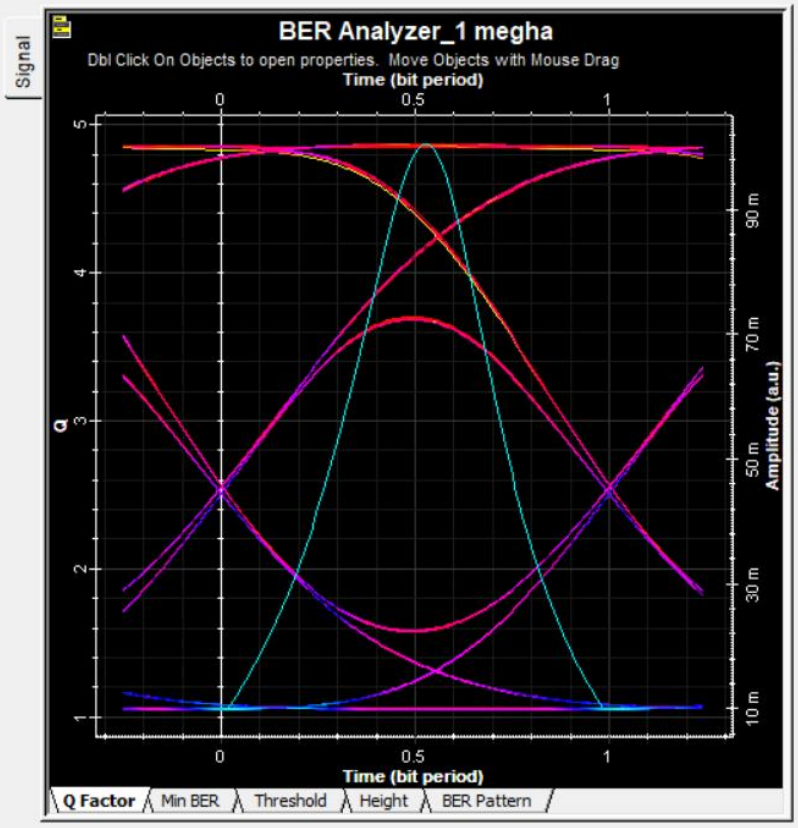
Test Case 5: Channel Power Variation

Channel Power	Min BER	Max Q-Factor
0 dBm		
5 dBm		
10 dBm		


OptiSystem - [DWDM exp 14]
File Edit View Layout Tools Report Script Add-Ins Window Help
Layout 1
Sweep Iteration 1
Layout: Layout 1
Author:
WDM Analyzer
Lower frequency limit = 185 THz
Upper frequency limit = 200 THz
Optical Fiber
Length = 100 km
EDFA
Length = 5 m
Optical Fiber 1
Length = 100 km
EDFA 1
Length = 5 m
Tx
WDM Transmitter
Frequency = 193.1 THz
Frequency spacing = 100 GHz
Power = 20 dBm
Bit rate = Bit rate Bits/s
Modulation type = NRZ
WDM Mux 4x1
Loop Control
Number of loops = 0
WDM Demux 1x4
Optical Power Meter
Optical Spectrum Analyzer
Rx
Optical Receiver
Cutoff frequency = 0.75 * Bit rate Hz
BER Analyzer megha
Optical Receiver 1
BER Analyzer_1 megha
Optical Receiver 2
BER Analyzer_2 megha
Optical Receiver 3
BER Analyzer_3 megha


BER Analyzer
Signal
BER Analyzer megha
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Q
Amplitude (a.u.)
100 m
80 m
60 m
40 m
20 m
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 4.88337
Min. BER 4.84975e-007
Eye Height 0.0270268
Threshold 0.0380388
Decision Inst. 0.523438
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008

BER Analyzer



Signal Index: 0

Auto Set

Show Eye Diagram

Analysis

Max. Q Factor	4.86909
Min. BER	5.1474e-007
Eye Height	0.0268445
Threshold	0.037942
Decision Inst.	0.523438

Invert Colors

Color Grade

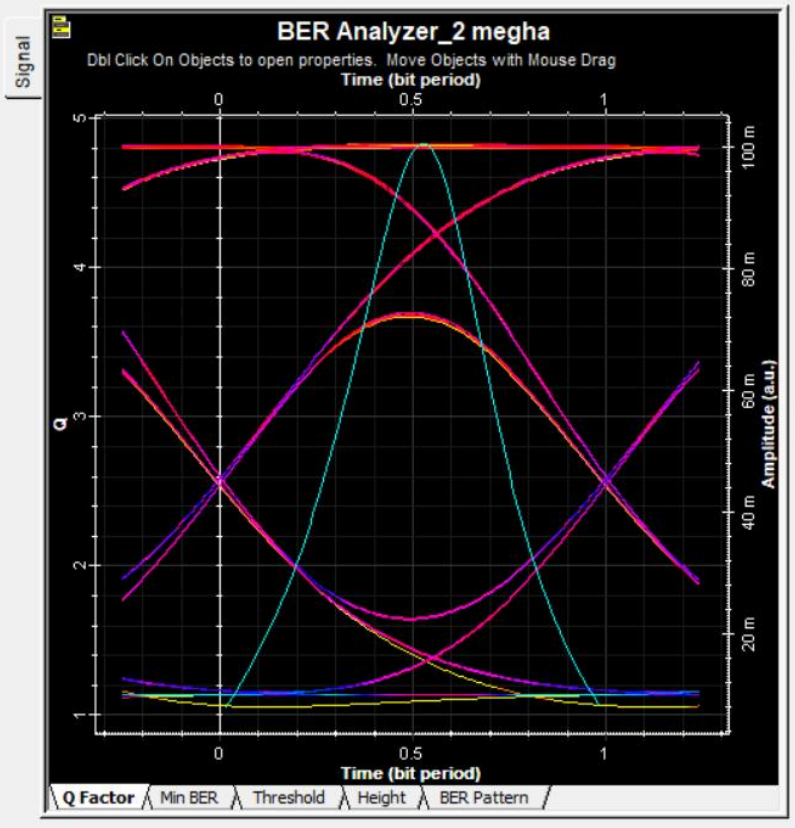
Patterns

Calculate Patterns

Patterns

Pattern 1	1e-012
Pattern 2	1e-011
Pattern 3	1e-010
Pattern 4	1e-009
Pattern 5	1e-008

BER Analyzer



Signal Index: 0

Auto Set

Show Eye Diagram

Analysis

Max. Q Factor	4.82701
Min. BER	6.48168e-007
Eye Height	0.0265288
Threshold	0.0385497
Decision Inst.	0.523438

Invert Colors

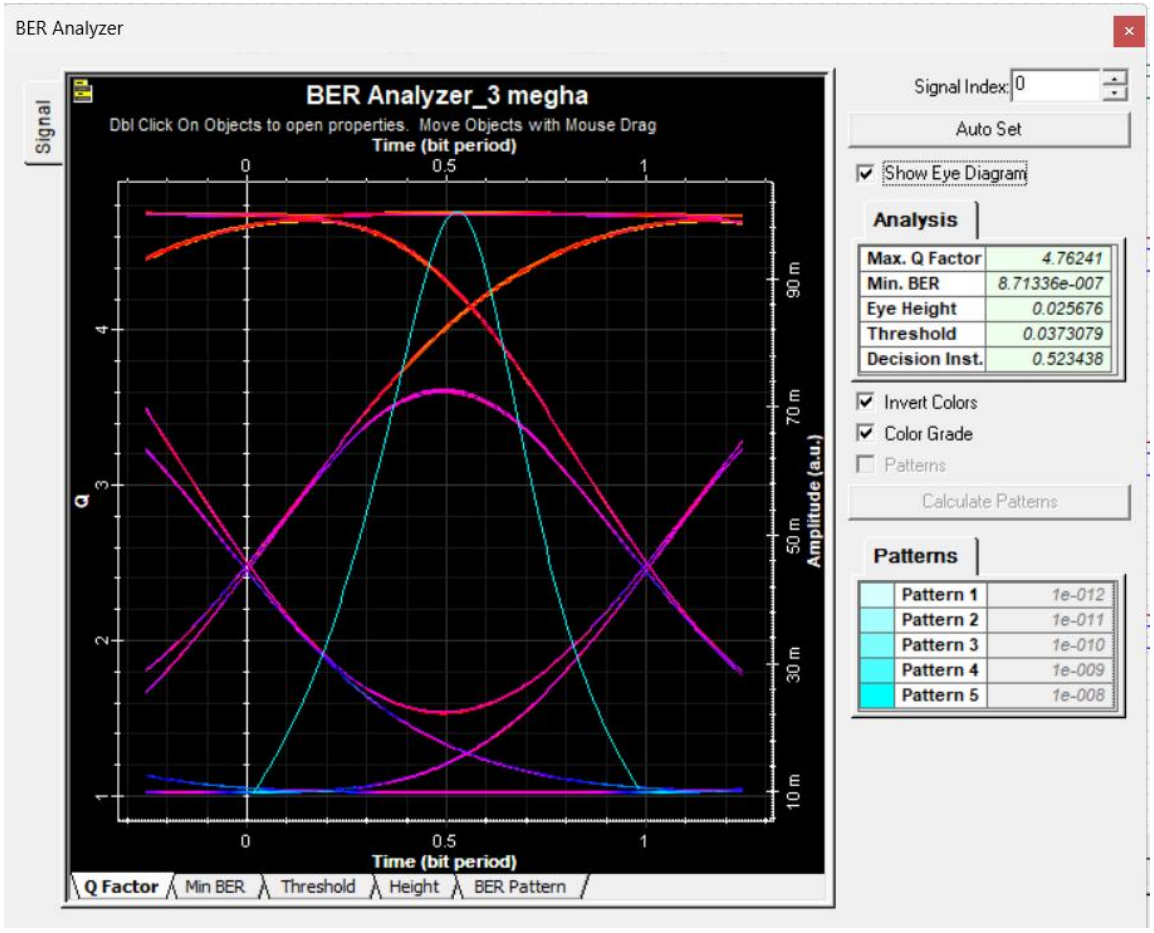
Color Grade

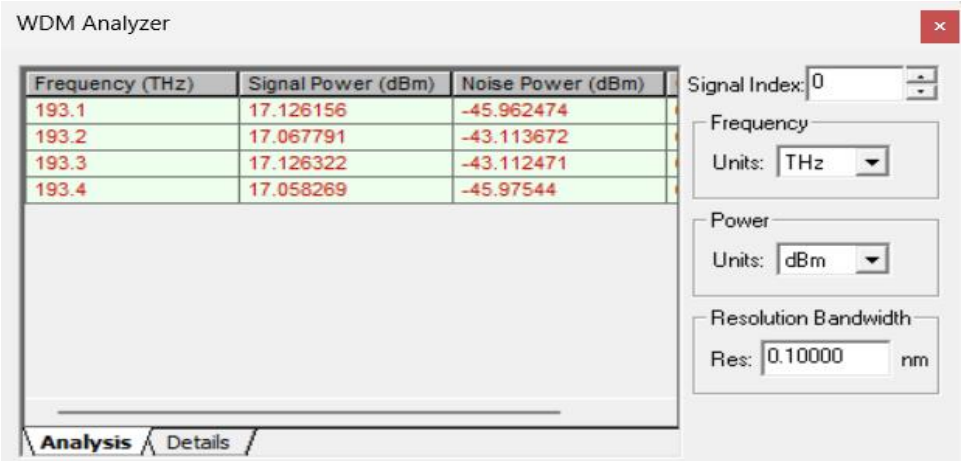
Patterns

Calculate Patterns

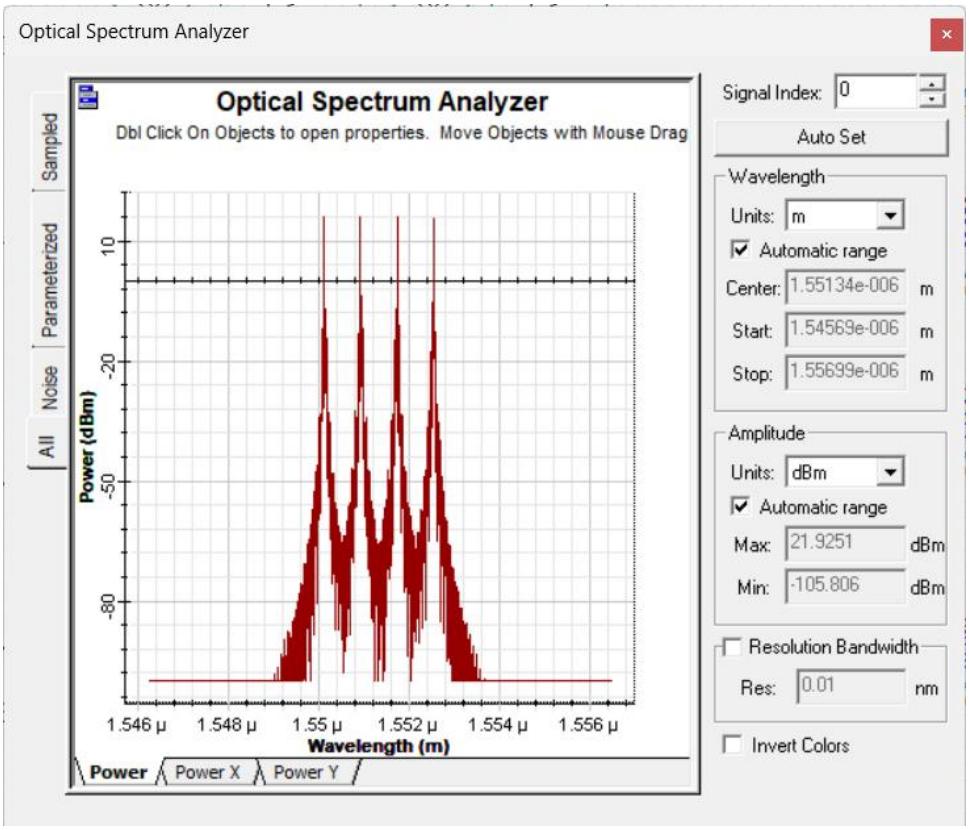
Patterns

Pattern 1	1e-012
Pattern 2	1e-011
Pattern 3	1e-010
Pattern 4	1e-009
Pattern 5	1e-008


BER Analyzer
Signal
BER Analyzer_3 megha
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Q
Amplitude (a.u.)
90 m
70 m
50 m
30 m
10 m
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 4.76241
Min. BER 8.71336e-007
Eye Height 0.025676
Threshold 0.0373079
Decision Inst. 0.523438
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008


WDM Analyzer
Frequency (THz) Signal Power (dBm) Noise Power (dBm)
193.1 17.126156 -45.962474
193.2 17.067791 -43.113672
193.3 17.126322 -43.112471
193.4 17.058269 -45.97544
Analysis
Details
Signal Index: 0
Frequency
Units: THz
Power
Units: dBm
Resolution Bandwidth
Res: 0.10000 nm


Optical Power Meter
205.621 E-3 W
23.131 dBm
Signal Index: 0
Total Power



Student Reflection:

Through this experiment, I understood how Dense Wavelength Division Multiplexing (DWDM) enables multiple optical signals to transmit simultaneously through a single optical fiber using different wavelengths. I learned how wavelength spacing, transmission distance, modulation format, channel power, and optical amplifier placement affect system performance. By analyzing the BER, Q-factor, eye diagrams, and optical spectrum using OptiSystem, I understood how to evaluate signal quality and transmission reliability. This experiment helped me relate theoretical concepts of DWDM to practical high-capacity optical communication systems.

Result:

Thus, the performance of a DWDM system was evaluated successfully using OptiSystem. The effects of wavelength spacing, transmission distance, modulation format, channel power, and amplifier placement on BER and Q-factor were analyzed using simulation results. The eye diagrams and optical spectrum helped assess signal quality and understand the factors affecting reliable long-distance optical communication.